

P8107 FINAL EXAM

Fall 2022

Name:

UNI:

For this exam you allowed up to four sheets of paper (front and back). No other reference materials may be used. You have 1 hour and 30 minutes to complete this exam. Be sure to show all your work. You may attach additional pages to your exam paper if needed.

Honor code:

I have not and will not give or receive aid in this examination nor have I concealed any violation of the University Honor Code.

Sign here: _____

1. (20 points) Random variables $Y_1, Y_2, \dots, Y_n$ have a pdf $f(y) = \frac{1}{\theta} y^{\frac{1-\theta}{\theta}}$. Find the maximum likelihood estimator of θ .

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} y_i^{\frac{1-\theta}{\theta}} = \frac{1}{\theta^n} \prod_{i=1}^n y_i^{\frac{1-\theta}{\theta}}$$

$$\begin{aligned} \ell(\theta) &= \log L(\theta) = -n \log \theta + \sum_{i=1}^n \frac{1-\theta}{\theta} \log y_i \\ &= -n \log \theta + \frac{1-\theta}{\theta} \sum \log y_i \end{aligned}$$

QUOTIENT RULE

$$\begin{aligned} \frac{d}{d\theta} \ell(\theta) &= -\frac{n}{\theta} + (\sum \log y_i) \frac{\theta(-1) - (1-\theta)(1)}{\theta^2} \\ &= -\frac{n}{\theta} + (\sum \log y_i) \frac{-\theta - 1 + \theta}{\theta^2} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum \log y_i \end{aligned}$$

set = 0 solve for θ :

$$\frac{n}{\theta} = -\frac{1}{\theta^2} \sum \log y_i \quad \theta = -\frac{\sum \log y_i}{n}$$

"CANDIDATE" MLE

$$\frac{d^2}{d\theta^2} \ell(\theta) = \frac{n}{\theta^2} + \frac{2}{\theta^3} \sum \log y_i = \frac{1}{\theta^2} \left(n + \frac{2 \sum \log y_i}{\theta} \right)$$

check to see whether 2nd derivative evaluated at

$\hat{\theta} = -\frac{\sum \log y_i}{n}$ is negative

$$\begin{aligned} \left. \frac{d^2}{d\theta^2} \ell(\theta) \right|_{\theta = -\frac{\sum \log y_i}{n}} &= \left(\frac{n}{\sum \log y_i} \right)^2 \left(n + \frac{2 \sum \log y_i}{-\frac{\sum \log y_i}{n}} \right) \\ &= \frac{n^2}{(\sum \log y_i)^2} (n - 2n) = -\frac{n}{(\sum \log y_i)^2} \end{aligned}$$

This is < 0 so this indeed is MLE

$$\hat{\theta} = -\frac{\sum \log y_i}{n}$$

2. (10 points) A hypothesis testing procedure for $H_0: \mu = 18$ vs. $H_a: \mu > 18$ is to reject H_0 if $\bar{Y} > 19$. Calculate the power of this test for $n = 25$ and $\sigma^2 = 9$ if the true mean is 20. (For this problem it is okay to leave your answer in terms of the cdf of a standard normal distribution, e.g., $P(Z < 1.5)$.)

$$RR: \{ \bar{Y} : \bar{Y} > 19 \}$$

Want $P(\bar{Y} > 19)$ when $\mu = 20$ (and $n = 25, \sigma^2 = 9$)

$$P(\bar{Y} > 19) = P\left(\frac{\bar{Y} - 20}{\sqrt{9/25}} > \frac{19 - 20}{\sqrt{9/25}}\right) = P\left(Z > -\frac{1}{\frac{3}{5}}\right)$$

$$= P\left(Z > -\frac{5}{3}\right)$$



3. (20 points) To test $H_0: \mu_X - \mu_Y = 2$ vs. $H_a: \mu_X - \mu_Y \neq 2$ you will find that the likelihood ratio test is equivalent to rejecting H_0 if $\bar{X} - \bar{Y}$ is too "large" or too "small", i.e., reject H_0 if $\bar{X} - \bar{Y} < K_1$ or if $\bar{X} - \bar{Y} > K_2$. Assuming that $X_1, X_2, \dots, X_m$ are iid $N(\mu_X, 1)$ and $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu_Y, 1)$ (and that the X observations are independent of the Y observations), find K_1 and K_2 so that the probability of a Type I error is α .

Choose K_1 so that $P(\bar{X} - \bar{Y} < K_1) = \frac{\alpha}{2}$ if H_0 true

Choose K_2 so that $P(\bar{X} - \bar{Y} > K_2) = \frac{\alpha}{2}$ if H_0 true

$$\bar{X} \sim N(\mu_X, \frac{1}{m}) \quad \bar{Y} \sim N(\mu_Y, \frac{1}{n})$$

$$\bar{X} - \bar{Y} \sim N(\mu_X - \mu_Y, \frac{1}{m} + \frac{1}{n})$$

$$\text{If } H_0 \text{ is true, } \bar{X} - \bar{Y} \sim N(2, \frac{1}{m} + \frac{1}{n})$$

Assuming H_0 is true:

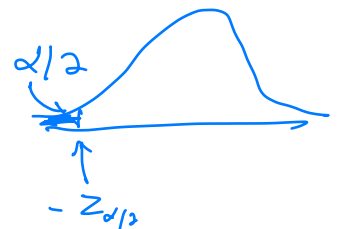
$$P(\bar{X} - \bar{Y} < K_1) = P\left(\frac{\bar{X} - \bar{Y} - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}} < \frac{K_1 - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}}\right) \stackrel{\text{set}}{=} \frac{\alpha}{2}$$

$$P\left(Z < \frac{K_1 - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}}\right) = \frac{\alpha}{2}$$

So (by definition)

$$\frac{K_1 - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}} = -z_{\alpha/2}$$

$$K_1 = 2 - \sqrt{\frac{1}{m} + \frac{1}{n}} z_{\alpha/2}$$



($z_{.025} = 1.96$,
for example)

$$P(\bar{X} - \bar{Y} > K_2) \stackrel{\text{set}}{=} \frac{\alpha}{2}$$

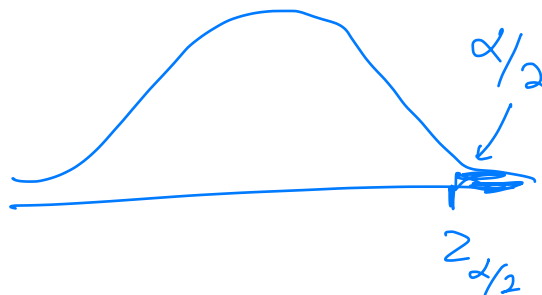
$$P\left(\frac{\bar{X} - \bar{Y} - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}} > \frac{K_2 - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}}\right) = \frac{\alpha}{2}$$

$$P(Z > \frac{K_2 - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}}) = \frac{\alpha}{2}$$

So (by definition)

$$\frac{K_2 - 2}{\sqrt{\frac{1}{m} + \frac{1}{n}}} = Z_{\alpha/2}$$

$$K_2 = 2 + Z_{\alpha/2} \sqrt{\frac{1}{m} + \frac{1}{n}}$$



4. (25 points) Random variables $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown.

- The natural estimator of μ is $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. Calculate the variance of $\bar{Y}$.
- A colleague thinks that μ is probably about 10 and suggests an alternative estimator which consists of adding an extra "observation" that is equal to 10, i.e., $\hat{\mu}^* = \frac{1}{n+1} [(\sum_{i=1}^n Y_i) + 10]$. Your colleague claims that this estimator will have smaller variance than $\bar{Y}$. Calculate the variance of $\hat{\mu}^*$ and compare it with that of $\bar{Y}$. Is your colleague correct?
- Calculate the MSE of both estimators.
- Which estimator has better MSE if in fact $\mu = 10$?
- Which estimator has better MSE if in fact $\mu = 9$ and $n = 4$?

$$(a) \text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n} \sum Y_i\right) = \frac{1}{n^2} \text{Var}\left(\sum Y_i\right) = \frac{1}{n^2} \sum \text{Var}(Y_i) \\ = \frac{1}{n^2} \sum \sigma^2 = \frac{1}{n^2} \cdot n \sigma^2 = \frac{\sigma^2}{n}$$

$$(b) \text{Var}(\hat{\mu}^*) = \text{Var}\left(\frac{1}{n+1} (\sum Y_i + 10)\right) = \frac{1}{(n+1)^2} \text{Var}(\sum Y_i + 10) \\ = \frac{1}{(n+1)^2} (n\sigma^2 + 0) \quad \text{Var of a constant} = 0$$

$$= \frac{n}{(n+1)^2} \sigma^2$$

$$(c) E[\bar{Y}] = E\left[\frac{1}{n} \sum Y_i\right] = \frac{1}{n} \sum E[Y_i] = \frac{1}{n} \cdot n \mu = \mu$$

$$\text{So bias}(\bar{Y}) = E[\bar{Y}] - \mu = \mu - \mu = 0$$

$$\text{MSE}(\bar{Y}) = \text{bias}^2(\bar{Y}) + \text{Var}(\bar{Y}) = 0 + \frac{\sigma^2}{n} = \frac{\sigma^2}{n}$$

$$E[\hat{\mu}^*] = E\left[\frac{1}{n+1} (\sum Y_i + 10)\right] = \frac{1}{n+1} (\sum E[Y_i] + E[10]) \\ = \frac{1}{n+1} (n\mu + 10) = \frac{n}{n+1} \mu + \frac{10}{n+1}$$

$$\text{So bias}(\hat{\mu}^*) = E[\hat{\mu}^*] - \mu = \frac{n}{n+1} \mu + \frac{10}{n+1} - \mu$$

$$= \frac{\cancel{n}\mu + 10 - \cancel{n}\mu - \mu}{n+1} = \frac{10 - \mu}{n+1}$$

$$\text{So } \text{MSE}(\hat{\mu}^*) = \frac{(10-\mu)^2}{(n+1)^2} + \frac{n\sigma^2}{(n+1)^2}$$

$$(d) \text{ If } \mu=10 \quad \text{MSE}(\bar{Y}) = \frac{\sigma^2}{n}$$

$$\text{MSE}(\hat{\mu}^*) = 0 + \frac{n\sigma^2}{(n+1)^2}$$

$$\frac{\text{MSE}(\bar{Y})}{\text{MSE}(\hat{\mu}^*)} = \frac{\frac{\cancel{\sigma^2}}{n}}{\frac{n\cancel{\sigma^2}}{(n+1)^2}} = \frac{(n+1)^2}{n^2}$$

this is > 1 for all n so if $\mu=10$, $\hat{\mu}^*$ has better MSE than $\bar{Y}$

$$(e) \text{ If } \mu=9 \text{ and } n=4$$

$$\text{MSE}(\bar{Y}) = \frac{\sigma^2}{4}$$

$$\text{MSE}(\hat{\mu}^*) = \frac{1^2}{5^2} + \frac{4\sigma^2}{5^2}$$

$$= \frac{4}{25}\sigma^2 + \frac{1}{25}$$

$$\text{Check: when is } \frac{4}{25}\sigma^2 + \frac{1}{25} < \frac{1}{4}\sigma^2 \quad ?$$

$$\text{when } \frac{1}{25} < \sigma^2 \left(\frac{25-16}{100} \right)$$

5. (10 points) For observations $Y_1, Y_2, \dots, Y_n$ that are iid $N(\mu, \sigma^2)$, the maximum likelihood estimator of σ^2 is $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$.

a. Find $E[\hat{\sigma}^2]$ and $Var(\hat{\sigma}^2)$.

b. Based on (a), is $\hat{\sigma}^2$ a consistent estimator of σ^2 ? Explain why or why not.

a) We know $\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$ (Here $s^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$)

So $E\left[\frac{(n-1)s^2}{\sigma^2}\right] = n-1 \Rightarrow \frac{n-1}{\sigma^2} E[s^2] = n-1$
 (because Exp of χ^2) $\Rightarrow$ So $E[s^2] = \sigma^2$

But $\hat{\sigma}^2 = \frac{n-1}{n} \frac{1}{n-1} \sum (Y_i - \bar{Y})^2 = \frac{n-1}{n} s^2$

So $E[\hat{\sigma}^2] = E\left[\frac{n-1}{n} s^2\right] = \frac{n-1}{n} E[s^2]$
 $= \boxed{\frac{n-1}{n} \sigma^2}$

Var of $\chi^2 \Rightarrow$
 $Var\left(\frac{(n-1)s^2}{\sigma^2}\right) = 2(n-1)$

$\frac{(n-1)^2}{\sigma^4} Var(s^2) = 2(n-1)$

$Var(s^2) = \frac{2\sigma^4}{n-1}$

So $Var(\hat{\sigma}^2) = Var\left(\frac{n-1}{n} s^2\right) = \frac{(n-1)^2}{n^2} Var(s^2)$
 $= \frac{(n-1)^2}{n^2} \frac{2\sigma^4}{n-1} = \boxed{\frac{2(n-1)\sigma^4}{n^2}}$

$$(b) \quad E[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2 = \sigma^2 - \frac{1}{n} \sigma^2 \rightarrow \sigma^2 \quad \text{as } n \rightarrow \infty$$

$$\text{Var}(\hat{\sigma}^2) = \frac{2(n-1)\sigma^4}{n^2} = \frac{2n\sigma^4}{n^2} - \frac{2\sigma^4}{n^2}$$

$$= \frac{2\sigma^4}{n} - \frac{2\sigma^4}{n^2}$$

$\searrow \quad \searrow$
 $0 \quad 0$

Since $\lim_{n \rightarrow \infty} E[\hat{\sigma}^2] = \sigma^2$

and $\lim_{n \rightarrow \infty} \text{Var}(\hat{\sigma}^2) = 0,$

$\hat{\sigma}^2$ is a consistent estimator of σ^2 by Theorem.

6. (15 points) To estimate the probability p given a binomial random variable $Y \sim \text{binom}(n, p)$, the maximum likelihood estimator can be found to be $\hat{p} = \frac{Y}{n}$. Using the large-sample properties of MLEs, derive an approximate 95% confidence interval for p . (To receive credit for this problem, you must start with the expression for $\sigma_{\hat{\theta}}^2$ and derive the interval – not just write down the answer.)

$$\sigma_{\hat{\theta}}^2 = \frac{1}{n E \left[-\frac{\partial^2}{\partial \theta^2} \log f(Y|\theta) \right]}$$

general formula

NOTE: This n is the number of random variables observed, as in $Y_1, \dots, Y_n$. In

$$\sigma_{\hat{p}}^2 = \frac{1}{n E \left[-\frac{d^2}{dp^2} \log f(Y|p) \right]}$$

$$f(Y|p) = \binom{n}{Y} p^Y (1-p)^{n-Y}$$

$$\log f(Y|p) = \log \binom{n}{Y} + Y \log p + (n-Y) \log (1-p)$$

only one obs - Y

$$\frac{d}{dp} \log f(Y|p) = \frac{Y}{p} - \frac{n-Y}{1-p}$$

So this $\frac{d^2}{dp^2} \log f(Y|p) = -\frac{Y}{p^2} - \frac{n-Y}{(1-p)^2}$

n is just 1

$$E \left[\frac{d^2}{dp^2} \log f(Y|p) \right] = -\frac{np}{p^2} - \frac{n-np}{(1-p)^2}$$

$$= -\frac{n}{p} - \frac{n(1-p)}{(1-p)^2} = -n \left(\frac{1}{p} + \frac{1}{1-p} \right)$$

$$= -n \left(\frac{1-p+p}{p(1-p)} \right) = -\frac{n}{p(1-p)}$$

$$\text{So } \hat{\sigma}_{\hat{p}}^2 = \frac{1}{-\left(-\frac{n}{p(1-p)}\right)} = \frac{p(1-p)}{n}$$

$$\text{So we know } \hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$$

So the interval is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

plug in $\hat{p}$ for p :

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$